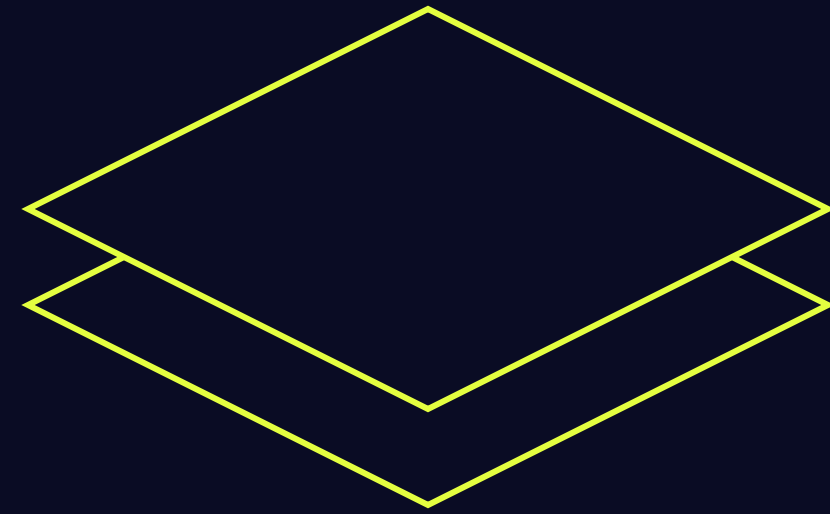
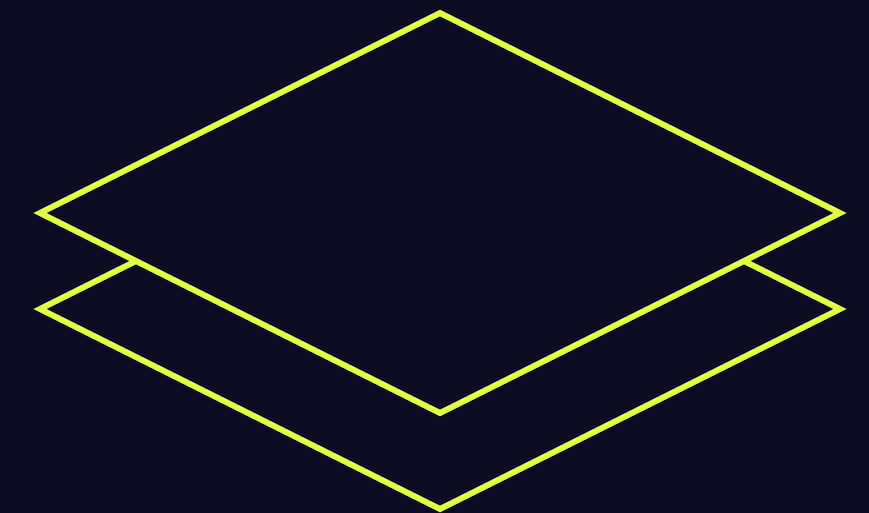




Scalable Data Architecture, APIs, and Storage Strategies of Healthcare System

Kuliah Tamu · Teknologi Kedokteran ITS



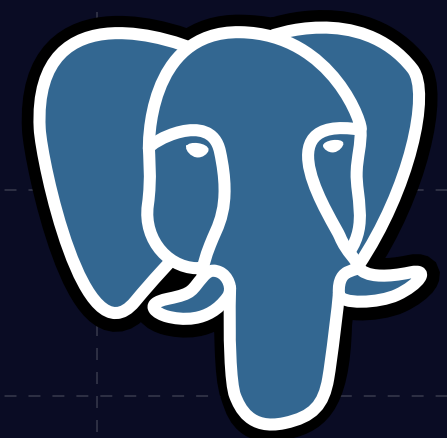
Contents

[01] Introduction

[02] Building systems

[03] Building yourself

Tech Stacks



Disclaimer

[02]

YOUR WORDS never built a hospital system

Disuruh bikin sistem klinik.
Mulai dari mana?

Bikin yang jalan dulu.

Gedein pas udah kerasa sempit.

Gimana tau udah sempit?

[02]

Latency

The time it takes to service a request.

Traffic

A measure of how much demand is being placed on your system

Errors

The rate of requests that fail

Saturation

How “full” your service is.

Latency

[02]

INDICATOR

$\leq 0.8s$ TTFB: good

$> 1.8s$ TTFB: poor

$> 10s$ the limit for keeping the user's attention
focused on the dialogue

REMIEDIATION

| It's usually wise to set a deadline.

| Serve lower-quality, cheaper-to-compute
results to the user.

Traffic

[02]

INDICATOR

> $\frac{1}{2}$ of load-tested max req/s

can your service properly handle double the traffic

REMEDIATION

limiting the volume of requests by criteria such as IP address

Good capacity planning can reduce the probability that a cascading failure will occur.

Errors

[02]

INDICATOR

99.9% SLO → error budget 0.1%

> 1.44% errors for 1 hour → page (burn rate 14.4)

> 0.1% errors for 3 days → ticket (burn rate 1)

REMEDIATION

Always use randomized exponential backoff when scheduling retries.

Limit retries per request. Don't retry a given request indefinitely.

Saturation

[02]

INDICATOR

< 5% disk free → warning

< 3% disk free → critical

≤ 4h until disk full (predicted) → critical

> 90% CPU / memory

REMIEDIATION

Load shedding drops some proportion of load by dropping traffic as the server approaches overload.

adding tasks can be the most expedient way to recover from the outage.

Di klinik, apa yang sempit duluan?

Latency



Traffic



Errors

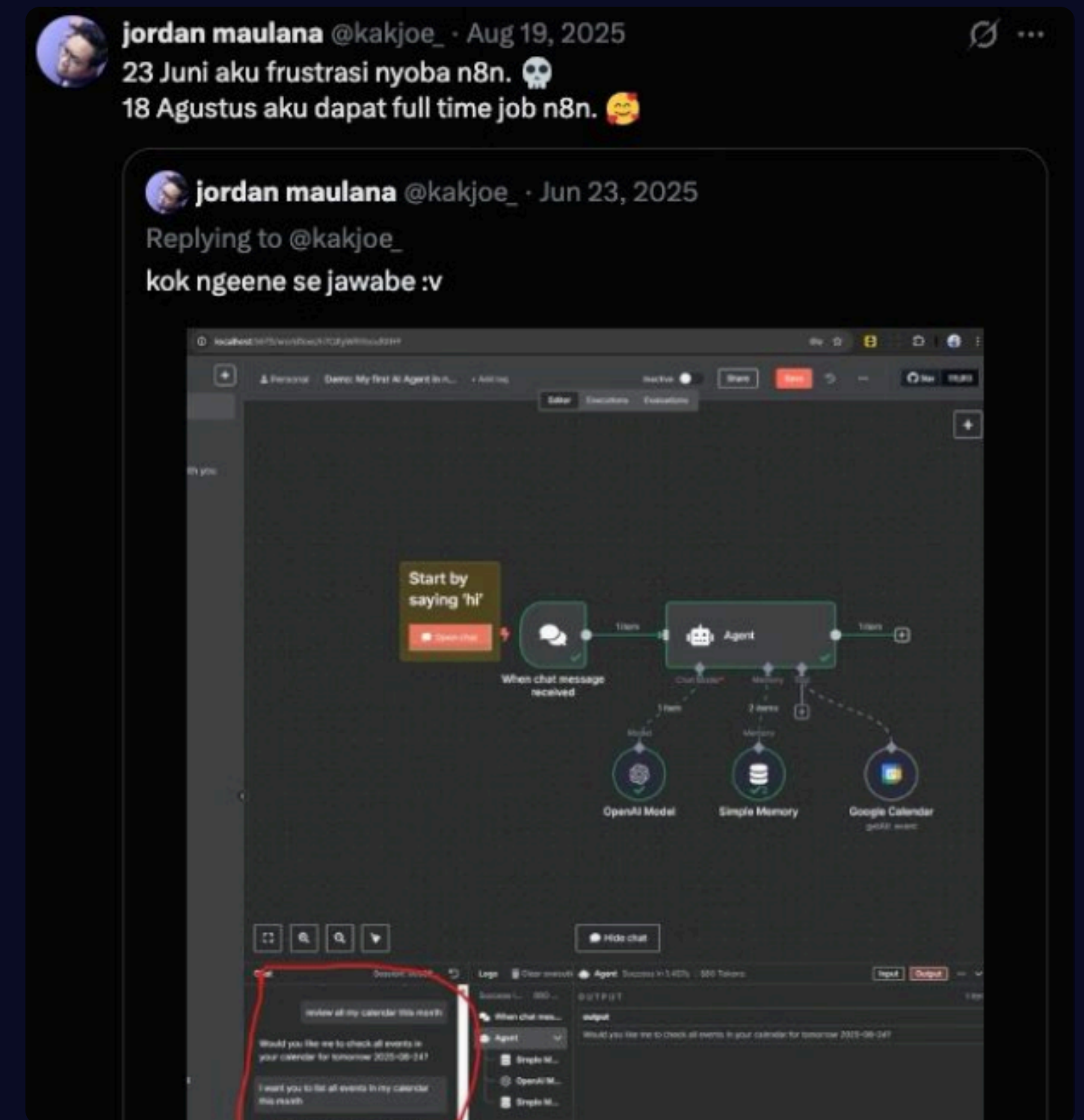


Saturation

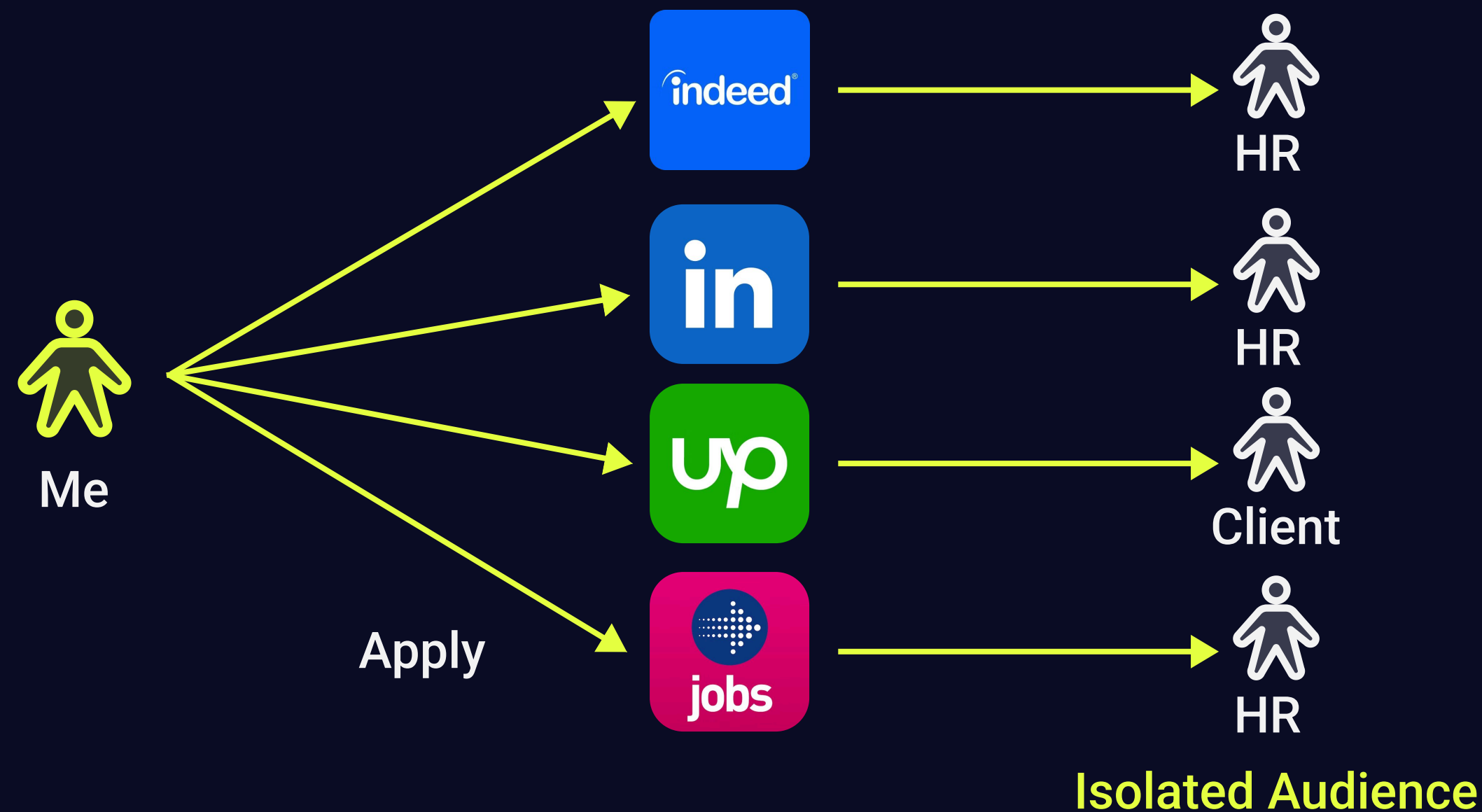


23 Juni aku frustrasi nyoba n8n.

18 Agustus aku dapat full time job n8n.



Challenge: Visibility



Publish Yourself

Make ourselves more interesting to be seen by anyone.

Aim:

- Reference to jobs & more people
- Collaboration
- Seminar invitation
- Projects

[02]

Creator



How to start

Donts

Cari niche di depan

Kelamaan mikir :v

Berharap viral

Do

Effortless first

Find joy

Share journey



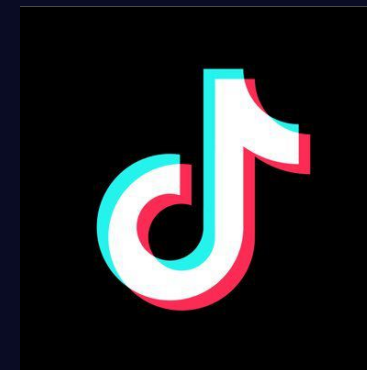
- Achievements
- Journey / progression
- Professional takes texts



- Personal takes short videos
- Carousel
- Personal journey (Day 1 apply loker)



- Personal takes texts
- Genuine open question
- Journey / progression



- Personal takes short videos
- Tiktok Live

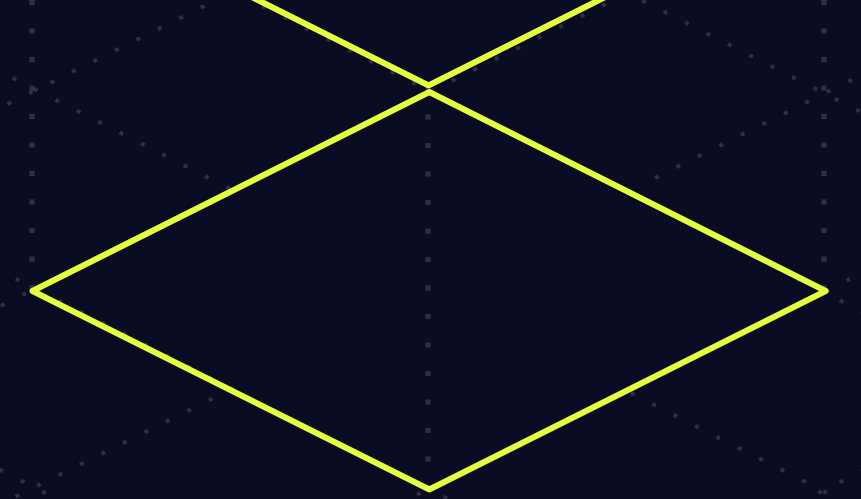
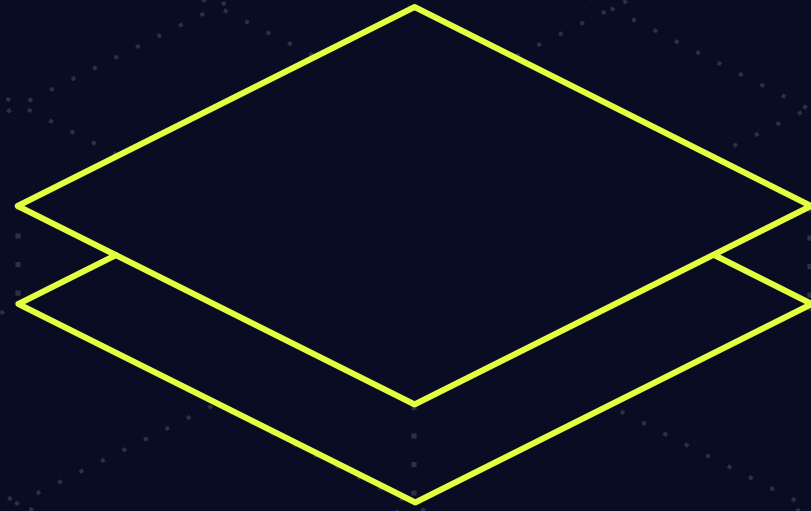


- Personal takes short videos
- Personal story long videos

Can I repeat the same result?

I don't know

Luck = Preparation x **Opportunity**



Now is your turn!

QnA

